FabrIQ

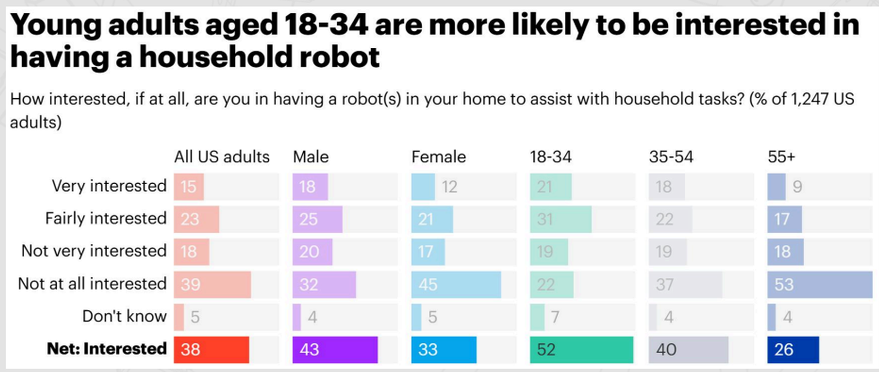
Laundry? Handled.

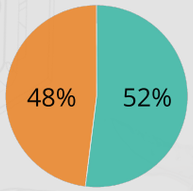
UNIVERSITY OF WATERLOO

Eric Wang, Jake Chua, Joshua Scripcaru, Kevin Wang, Phillip Rojkov

Motivation

- Textiles are hard to work with in robotics
  - Can be folded, obscuring themselves from vision
  - Difficult to model in simulations because they are not rigid
- People are generally interested in having some robotic assistance with household tasks





52%

of Americans aged 18-34 are interested

Objective

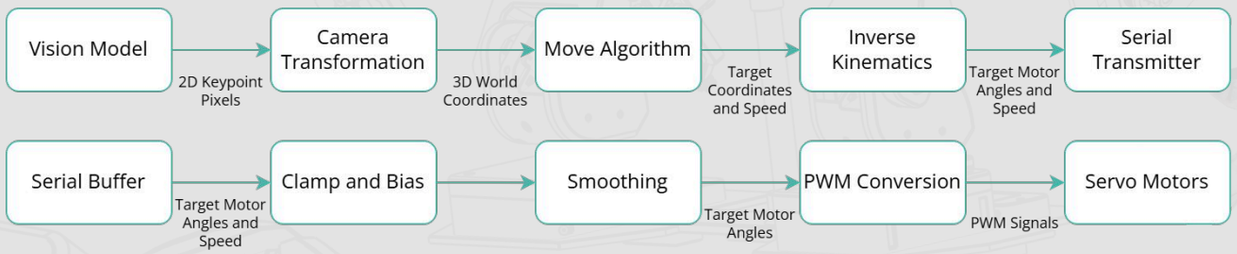
Construct a **generalized** robotic manipulation device which **visually** detects and automatically folds clothing

Features

- 6 DoF, 80cm effective range
- Customizable speed
- Easy-to-use calibration GUI
- Custom vision model
- Extreme low-cost (<$700)

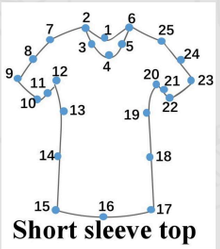
Design

Software Pipeline



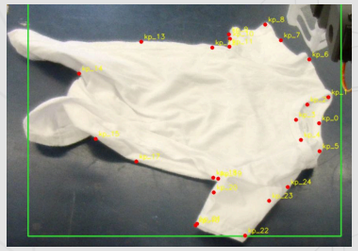
Vision Model

- Trained using **DeepFashion2** Dataset; a key-point labelled collection of shirts



Keypoint labels

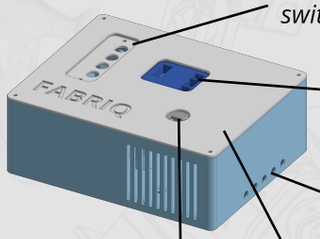
Rotated images

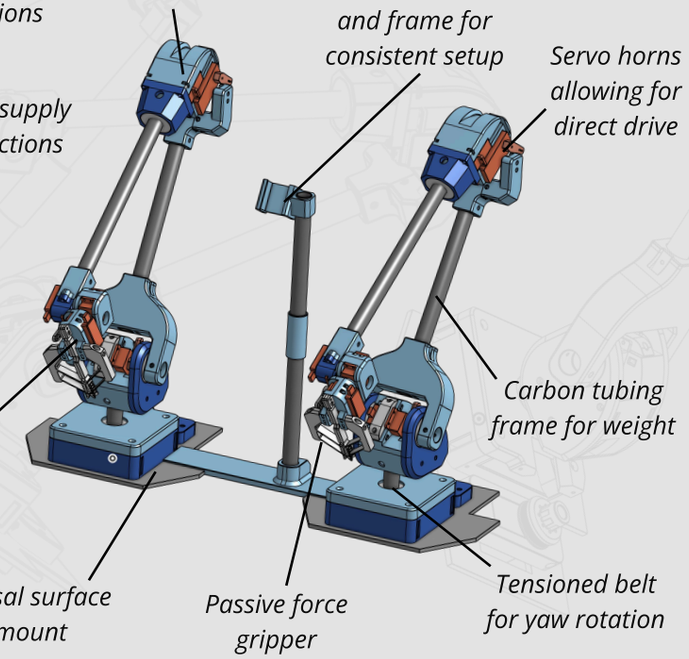


Predicted keypoints

- Used **Keypoint R-CNN**-based learning model to identify and obtain estimated keypoints
- Created custom data to further train vision model

Mechanical Design





References

Y. Ge, R. Zhong, X. Wang, X. Tang, and P. Luo, "DeepFashion2," A Versatile Benchmark for Detection, Pose Estimation, Segmentation and Re-Identification of Clothing Images, 2019. Accessed: 2026. [Online]. Available: https://github.com/switchablenorms/DeepFashion2